

# Amir Mirzai Golpayegani

📍 Calgary, AB, Canada

✉ [amirgolpaa24@gmail.com](mailto:amirgolpaa24@gmail.com) | ✉ [amir.mirzaigolpayega@ucalgary.ca](mailto:amir.mirzaigolpayega@ucalgary.ca) | ☎ +1 (587) 439-0436  
🌐 [linkedin.com/in/amir-golpayegani-218148165](https://www.linkedin.com/in/amir-golpayegani-218148165) | 🐙 [github.com/amirgolpaa24](https://github.com/amirgolpaa24) | 🌐 Portfolio

## Summary

M.Sc. Computer Science graduate with experience building Python/C++ software for machine learning, backend APIs, data pipelines, and AI workflow automation. Skilled in LLM workflows, AI agents, PyTorch, Django REST APIs, PostgreSQL, Docker, Git, Linux, AWS, and scientific computing, with hands-on experience turning complex domain data into reliable software tools.

## Education

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|--------------------------------------------------------------------------------|----------------------|-----------------------------------------------|
| <b>M.Sc. in Computer Science</b><br><i>University of Calgary</i>               | GPA: <b>3.94/4.0</b> | Sep 2022 – Feb 2026<br><i>Calgary, Canada</i> |
| <b>B.Sc. in Computer Engineering</b><br><i>Sharif University of Technology</i> | GPA: <b>3.69/4.0</b> | Sep 2016 – Feb 2021<br><i>Tehran, Iran</i>    |

## Technical Skills

**Programming:** Python, C++, JavaScript, SQL  
**ML / AI:** PyTorch, Scikit-learn, TensorFlow, CNNs, LLM workflows, AI agents, RAG/evaluation workflows  
**Backend / Engineering:** Django, Node.js, REST APIs, PostgreSQL, Docker, Git, Linux, AWS, GitHub Actions  
**Data / Geospatial:** NumPy, Pandas, SciPy, Google Earth Engine API, Discrete Global Grid System (DGGS), GDAL, Rasterio, SNAP  
**Visualization / Automation:** Streamlit, Polyscope, OpenGL, GLSL, n8n, OpenClaw, SearXNG, Telegram API

## Work Experience

**Back-End Developer** Jun 2020 – Jul 2022  
*Asr Gooyesh Pardaz* *Tehran, Iran*

- Developed Django REST APIs for NLP, text-to-speech, and speech-to-text service plans backed by PostgreSQL.
- Built authentication features including registration, Google login, password reset, and email/phone verification.
- Implemented backend logic for payment-service integration and storing plan/payment records in a relational database.
- Contributed to a WebRTC-based chatbot system using GPT-style responses and FAQ data to answer client questions.

**Research Software Developer** Sep 2022 – Feb 2026  
*BigGeo / Mitacs, University of Calgary* *Calgary, Canada*

- Built C++/Python software for scalable processing, storage, retrieval, and evaluation of large scientific datasets.
- Automated SAR and optical imagery preprocessing using GDAL, Rasterio, SNAP, and satellite data APIs.
- Developed DGGS-based encoding pipelines to convert domain data into structured tensor files.
- Improved encoding performance with C++ multithreading, reducing 10-tensor processing from 233s to 26s.
- Documented system design, validation results, and performance tradeoffs for thesis, publication, and collaborator review.

**Deep Learning Teaching Assistant** Jul 2026  
*Neuromatch Academy* *Remote, United States*

- Mentored an international pod of ~15 students through Neuromatch Academy's intensive three-week Deep Learning course.
- Facilitated hands-on PyTorch tutorials spanning MLPs, CNNs, attention/transformers, diffusion generative models, and reinforcement learning.
- Guided project groups through end-to-end deep learning research projects, from question formulation and literature review to model implementation and final presentations.

**Teaching Assistant** Dec 2022 – Apr 2025  
*Department of Computer Science, University of Calgary* *Calgary, Canada*

- **Developing Big Data Applications:** Mentored +25 students on AWS-based data pipelines, cloud storage, debugging, and scalable data workflows.
- **Working with Data at Scale:** Supported +20 students with large-scale data processing, data pipelines, and high-volume data analysis.
- **Database Management Systems:** +75 Assisted students with SQL, database design, course projects, grading, and exam support.

## Research and Technical Projects

**Spatiotemporal Earth Observation Data System Using DGGS** Sep 2022 – Feb 2026  
*First-author publication | M.Sc. thesis | C++, Python, DGGS, Wavelets, B-splines, Time-Series*

- Designed a DGGS-based framework for real-time multiresolution management of satellite imagery.
- Developed a tensor-based storage model for direct access to arbitrary regions at multiple resolutions.
- Implemented a triangular wavelet scheme to support multiresolution retrieval without image-pyramid redundancy.

- Built reusable Python/C++ workflows for structured data generation, retrieval, and performance testing.
- Retrieved 105,489 DGGs cells per time-series query in 13.9s, compared with about 220s using global retrieval.

DEM Super-Resolution Framework

May 2024 – Feb 2026

Python, PyTorch, CNNs, Scientific Imaging, Google Earth Engine API, Polyscope, Data Processing and Visualization

- Trained deep ResNet models for 4× DEM super-resolution, reconstructing 30m elevation data from 120m inputs.
- Built a Streamlit pipeline using Rasterio to parallelize GEE DEM tile downloads and systematically collect and preprocess mountainous terrain data worldwide.
- Implemented PyTorch training and evaluation workflows with terrain-aware metrics such as RMSE, slope, TRI, TPI, and wavelet-based errors.
- Developed a Polyscope-based 3D visualization tool to render DEM tiles from multiple angles and compare reconstruction quality in 3D.

Agentic Job Search Automation Platform

May 2026 – Present

OpenClaw, n8n, LLMs, AI Agents, Docker, Workflow Automation, SearXNG, Telegram API

- Building an agentic AI system to search jobs, analyze job fit, and generate tailored application materials.
- Designed LLM-based workflows for resume/job matching, skill-gap analysis, tool orchestration, and application decision support.
- Integrated Google Sheets, Google Drive, and Telegram API to track jobs and send automated application alerts.
- Prototyped the initial workflow in n8n and Docker before migrating toward a more flexible OpenClaw-based agent system.

Persian Text Normalizer for NLP

Oct 2016 – Feb 2021

Python, NLP, Text Processing

- Developed a Python-based text normalization toolkit to standardize Persian text for downstream NLP pipelines.
- Implemented preprocessing rules for cleaning, formatting, and preparing Persian text data for language-processing tasks.

Publication

Amir Mirzai Golpayegani, Mahmudul Hasan, Faramarz F. Samavati. *Real-Time Multiresolution Management of Spatiotemporal Earth Observation Data Using DGGs*. Remote Sensing, 2025. [doi:10.3390/rs17040570](https://doi.org/10.3390/rs17040570)

Certificates

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|------------------------------------------------------------------------------------------------------------------|----------|
| Supervised Machine Learning: Regression and Classification,<br>DeepLearning.AI and Stanford University, Coursera | Oct 2024 |
| Applied Machine Learning in Python,<br>University of Michigan, Coursera                                          | Jul 2020 |

Languages

English: Full Professional Proficiency (TOEFL 115/120)  
Persian: Native Proficiency

Achievement

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Teaching Assistant Excellence Laureate<br>Department of Computer Science, University of Calgary — awarded for outstanding teaching support ( <a href="#">link</a> ) | 2024/2025 |
| Ranked 17th Nationwide in Iran University Entrance Exam ( <a href="#">Konkour</a> )<br>Top 0.001% among all participants                                            | May 2016  |